

SortSphere – User Guide

Download, Build & Run from Command Line (Windows)

Prerequisites

Install the following:

- Java Development Kit (JDK 17 or later)
- Git
- Windows Command Prompt (CMD)

Verify installation:

```
java -version  
javac -version  
git --version
```

Step 1 – Download the Source Code

Open Command Prompt.

Navigate to the folder where you want the project.

Example:

```
cd C:\Users\YourName\Documents
```

Clone the repository:

```
git clone https://github.com/sarlayash/Initial-SortSphere-sorting-game.git
```

Move into the project folder:

```
cd Initial-SortSphere-sorting-game
```

Step 2 – Verify Project Structure

Run:

```
dir
```

You should see something similar to:

```
README.md
```

```
src\
```

```
.gitignore
```

Inside **src**:

```
dir src
```

Expected:

```
SortSphereGame.java
```

Step 3 – Compile the Project

Create an output folder:

```
mkdir out
```

Compile:

```
javac -d out src\SortSphereGame.java
```

If compilation is successful, there will be **no errors**.

Step 4 – Run the Game

Execute:

```
java -cp out SortSphereGame
```

The SortSphere window will open.

Step 5 – Play the Game

1. Enter numbers

Example:

8,3,10,1,6,4

2. Select an algorithm
 - Bubble Sort
 - Selection Sort
 - Insertion Sort
 - Merge Sort
3. Click

Start Animation

or

Step

to visualize each comparison.

Step 6 – Earn Badges

Complete all four algorithms.

Progress example:

Progress: 4 / 4 algorithms mastered

You will unlock:

 Sorting Champion

Step 7 – Generate Certificate

Click

Save Certificate

Enter your name.

Choose a location.

A PNG certificate will be generated.

Example:

SortSphere-Certificate.png

Updating the Project

Open CMD.

Go to the project folder:

```
cd Initial-SortSphere-sorting-game
```

Download the latest changes:

```
git pull
```

Recompile:

```
javac -d out src\SortSphereGame.java
```

Run:

```
java -cp out SortSphereGame
```

If You Modify the Code

Compile again:

```
javac -d out src\SortSphereGame.java
```

Run:

```
java -cp out SortSphereGame
```

Common Errors

Java not found

'java' is not recognized...

Solution

Install the JDK and ensure Java is added to your system [PATH](#).

Compilation error

cannot find symbol

Solution

Check for typing mistakes or incomplete code changes, then compile again.

Git not found

'git' is not recognized...

Solution

Install Git for Windows and restart Command Prompt.

Repository already exists

destination path already exists

Use:

`cd Initial-SortSphere-sorting-game`

or clone into a different folder.

Quick Start (One-Minute Setup)

```
git clone https://github.com/sarlayash/Initial-SortSphere-sorting-game.git
```

```
cd Initial-SortSphere-sorting-game
```

```
mkdir out
```

```
javac -d out src\SortSphereGame.java
```

```
java -cp out SortSphereGame
```

Enjoy learning sorting algorithms visually with **SortSphere**!